Probability

Combination Based Examples of Probability

Counting Rules – Multiplication

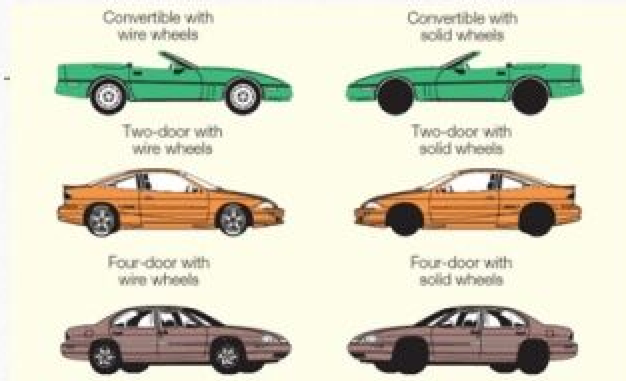
The multiplication formula indicates that if there are m ways of doing one thing and n ways of doing another thing, there are m x n ways of doing both.

Example: Dr. Delong has 10 shirts and 8 ties. How many shirt and tie outfits does he have?

(10)(8) = 80

Counting Rules – Multiplication: Example

An automobile dealer wants to advertise that for \$29,999 you can buy a convertible, a two-door sedan, or a four-door model with your choice of either wire wheel covers or solid wheel covers. How many different arrangements of models and wheel covers can the dealer offer?



MULTIPLICATION FORMULA Total number of arrangements $= (m)(n)$

We can employ the multiplication formula as a check (where m is the number of models and n the wheel cover type). From formula (5–8):

$$\text{Total possible arrangements} = (m)(n) = (3)(2) = 6$$

Counting Rules - Permutation

A permutation is any arrangement of r objects selected from n possible objects. The order of arrangement is important in permutations.

PERMUTATION FORMULA

$$_nP_r = \frac{n!}{(n-r)!}$$

where:

n is the total number of objects.

r is the number of objects selected.

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Counting - Combination

A combination is the number of ways to choose r objects from a group of n objects without regard to order.

COMBINATION FORMULA $${}_nC_r = \frac{n!}{r!(n-r)!}$$

where:

n is the total number of objects.

r is the number of objects selected.

Combination and Permutation Examples

COMBINATION EXAMPLE

There are 12 players on the Carolina Forest High School basketball team. Coach Thompson must pick five players among the twelve on the team to comprise the starting lineup. How many different groups are possible?

$$ {}_{12}C_5 = \frac{12!}{5!(12-5)!} = 792 $$

PERMUTATION EXAMPLE

Suppose that in addition to selecting the group, he must also rank each of the players in that starting lineup according to their ability.

$$ {}_{12}P_5 = \frac{12!}{(12-5)!} = 95{,}040 $$

Combination Based examples of Probability

- **Example 1:** A bag contains 6 white and 4 red and 10 black balls. 2 balls are drawn at random. What is the probability that both of them are black?

- **Solution:**
- Total number of balls = 6 + 4 + 10 = 20

2 balls can be drawn from 20 balls in 20_{C_2} ways = $\frac{20!}{2!\times(20-2)!}$

$= \frac{20\times19\times18!}{2\times18!} = 190\ ways$ (**Total No. of possible Events**)

Similarly, 2 ball from 10 black balls can be drawn in 10_{C_2} ways = $\frac{10!}{2!\times(10-2)!}$

$= \frac{10\times9\times8!}{2\times8!} = 45\ ways$ (No. of Favourable Events)

Therefore, the probability that two balls drawn are black = 45/190 = **0.24 Ans**

Combination Based examples of Probability

- **Example 2:** A bag contains 8 white and 4 red balls. 5 balls are drawn at random. What is the probability that 2 of them are red and 3 are white?

- **Solution:**
- Total number of balls = 8 + 4 = 12

5 balls can be drawn from 12 balls in 12_{C_5} ways $= \frac{12!}{5!\times(12-5)!}$

$= \frac{12\times11\times10\times9\times8\times7!}{5\times4\times3\times2\times7!} = 792\ ways$ (Total No. of possible Cases)

Similarly, 2 ball from 4 red balls can be drawn in 4_{C_2} ways $= \frac{4!}{2!\times(4-2)!}$

$= \frac{4\times3\times2!}{2\times2!} = 6\ ways$

Combination Based examples of Probability

Similarly, 3 ball from 8 white balls can be drawn in 8_{C_3} ways $= \frac{8!}{3!\times(8-3)!}$

$$= \frac{8\times7\times6\times5!}{3\times2\times5!} = 56\ ways$$

Therefore, the probability that two balls drawn are red and 3 black

$$\frac{\text{No. of Favourable Cases}}{\text{Total No. of possible Cases}} = \frac{4_{C_2} \times 8_{C_3}}{12_{C_5}}$$

$$= \frac{6\times56}{792} = \frac{14}{33} = \mathbf{0.424}\ \boldsymbol{Ans}$$

Probability

Addition Theorem

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Addition Theorem of Probability

- **Addition Theorem**: If two events A and B are mutually exclusive, the probability of occurrence of either A or B is the sum of the individual probability of A and B. Symbolically,
- $P(A \text{ or } B) = P(A) + P(B)$
- The addition theorem is also known as the theorem of total probability.
- **Proof of the Theorem**: If an event A can happen in $\boldsymbol{a_1}$ ways and B can happen in $\boldsymbol{a_2}$ ways, then the number of ways in which either event can happen is $\boldsymbol{a_1} + \boldsymbol{a_2}$.

Addition Theorem of Probability

- If total number of possible events is **n**, then by definition the probability of either first or the second event happening is

$$\frac{a_1 + a_2}{n} = \frac{a_1}{n} + \frac{a_2}{n}$$

But, $\frac{a_1}{n} = \mathrm{P(A)}$

and, $\frac{a_2}{n} = P(B)$

Hence, P(A or B) = P(A) + P(B),

The theorem can be extended to three or more mutually exclusive events.

Thus, P(A or B or C) = P(A) + P(B) + P(C). **Proved**

Addition Theorem of Probability

- **Example**: A bag contains 30 balls numbered from 1 to 30. One ball is drawn at random, find the probability that the number of the ball will be multiple of 5 or 9.

- **Solution**: Number of multiple of 5 (Event A) = (5, 10, 15, 20, 25 and 30) = 6

Number of multiple of 9 (Event B)= (9, 18, and 27) = 3

Total number of events = 30

$$P(A) = \frac{6}{30}$$

$$P(B) = \frac{3}{30}$$

$P(A \text{ or } B) = P(A) + P(B)$ (Since the events are mutually exclusive)

$$\frac{6}{30} + \frac{3}{30} = \frac{9}{30} = \frac{3}{10}\ \textit{Ans}$$

Addition Theorem of Probability

- **Example**: A person can hit a target in 3 out 4 shots, whereas another person can hit the target in 2 out of 3 shots. Find the probability of the targets being hit at all when they both try.
- **Solution:**

The probability that the first person hit the target = 3/4

The probability that the second person hit the target = 2/3

Addition Theorem of Probability

- The events are **not** mutually exclusive because both of them may hit the target. Hence,

P(A or B) = P(A) + P(B) – P(A and B)

$$= \left(\frac{3}{4} + \frac{2}{3}\right) - \left(\frac{3}{4} \times \frac{2}{3}\right)$$

$$= \frac{17}{12} - \frac{6}{12} = \frac{11}{12}\ Ans$$

Multiplication Theorem of Probability

- **Multiplication Theorem**: This theorem states that if two events A and B are independent the probability that they both will occur is equal to product of probability of their individual probabilities. Symbolically, if A and B are independent, then

  $\mathbf{P(A\ and\ B) = P(A)} \times P(B).$

  The theorem can be extended to three or more independent events. Thus,

  $\mathbf{P(A\ and\ B) = P(A)} \times P(B) \times P(C).$

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Multiplication Theorem of Probability

- **Example**: A bag contains 5 white and 3 black balls. Two balls are drawn at random one after another without replacement. Find the probability that
  (i) both are black; (ii) one black and one white.

- **Solution**:

(i) Probability of drawing a black ball in the first attempt $= \frac{3}{3+5} = \frac{3}{8}$

Probability of drawing a black ball in the second attempt $= \frac{2}{2+5} = \frac{2}{7}$

Probability that both balls drawn are black

$$P(A \text{ and } B) = P(A) \times P(B)$$

$$= \frac{3}{8} \times \frac{2}{7} = \frac{3}{28} \; Ans$$

Multiplication Theorem of Probability

- **Case I:** Probability of drawing a black ball in the first attempt $= \frac{3}{3+5} = \frac{3}{8}$

Probability of drawing a white ball in the second attempt $= \frac{5}{2+5} = \frac{5}{7}$

Probability that first ball is black and second is white

$$P(A\ and\ B) = P(A) \times P(B)$$

$$= \frac{3}{8} \times \frac{5}{7} = \frac{\mathbf{15}}{\mathbf{56}}\ \boldsymbol{Ans}$$

Multiplication Theorem of Probability

- **Case II:** Probability of drawing a white ball in the first attempt $= \frac{5}{3+5} = \frac{5}{8}$

Probability of drawing a black ball in the second attempt $= \frac{3}{3+4} = \frac{3}{7}$

Probability that first ball drawn is white and second black

$$P(A\ and\ B) = P(A) \times P(B)$$

$$= \frac{5}{8} \times \frac{3}{7} = \mathbf{\frac{15}{56}}$$

One ball is white and another black can happen in either way Case I or Case II.

Hence, probability that one ball is white and one is black $= \left(\frac{15}{56} + \frac{15}{56}\right) = \frac{30}{56} = \mathbf{\frac{15}{28}}$ **Ans**